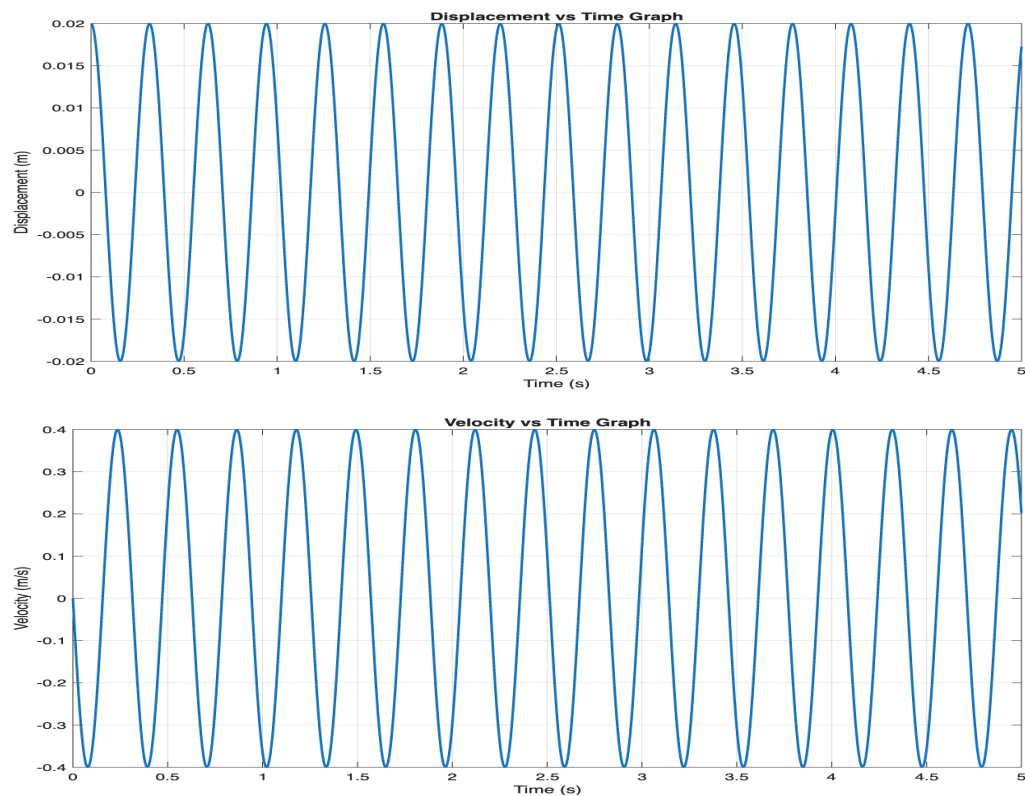


CODE

```
1 m = 5;           % Mass [kg]
2 k = 2000;        % Spring Stiffness [N/m]
3 x0 = 0.02;       % Initial Displacement [m]
4 v0 = 0;          % Initial Velocity [m/s]
5
6 Wn = sqrt(k/m);   % Natural Angular Frequency
7
8 t = linspace(0, 5, 1000);
9
10 x = x0 * cos(Wn * t); % Displacement Equation
11
12 v = -x0 * Wn * sin(Wn * t); % Velocity Equation
13
14 figure;          % Plot of the Displacement vs Time Graph
15 plot(t, x, 'LineWidth', 1.5);
16 ylabel('Displacement (m)');
17 xlabel('Time (s)');
18 title('Displacement vs Time Graph');
19 grid on;
20
21 figure;          % Plot of the Velocity vs Time Graph
22 plot(t, v, 'LineWidth', 1.5);
23 ylabel('Velocity (m/s)');
24 xlabel('Time (s)');
25 title('Velocity vs Time Graph');
26 grid on;
```

GRAPHS



COMMENTS

Nature of Motion – It is a sinusoidal, a simple harmonic motion. This is characterized by an exchange in potential energy of the spring and the kinetic energy of the mass.

Effect of Zero Damping – The energy does not dissipate. Constantly oscillates with a constant amplitude.

Relationship between displacement and velocity – When the displacement is at maximum, the velocity is zero and when the displacement is at zero, the velocity is at maximum.